

NCP-AIN AI Fabric Pro Mega Lab - Topology 3 Wall Poster

Large-format Fabric and Host / Edge zones with bigger labels for printing and tablet study

Fabric Zone

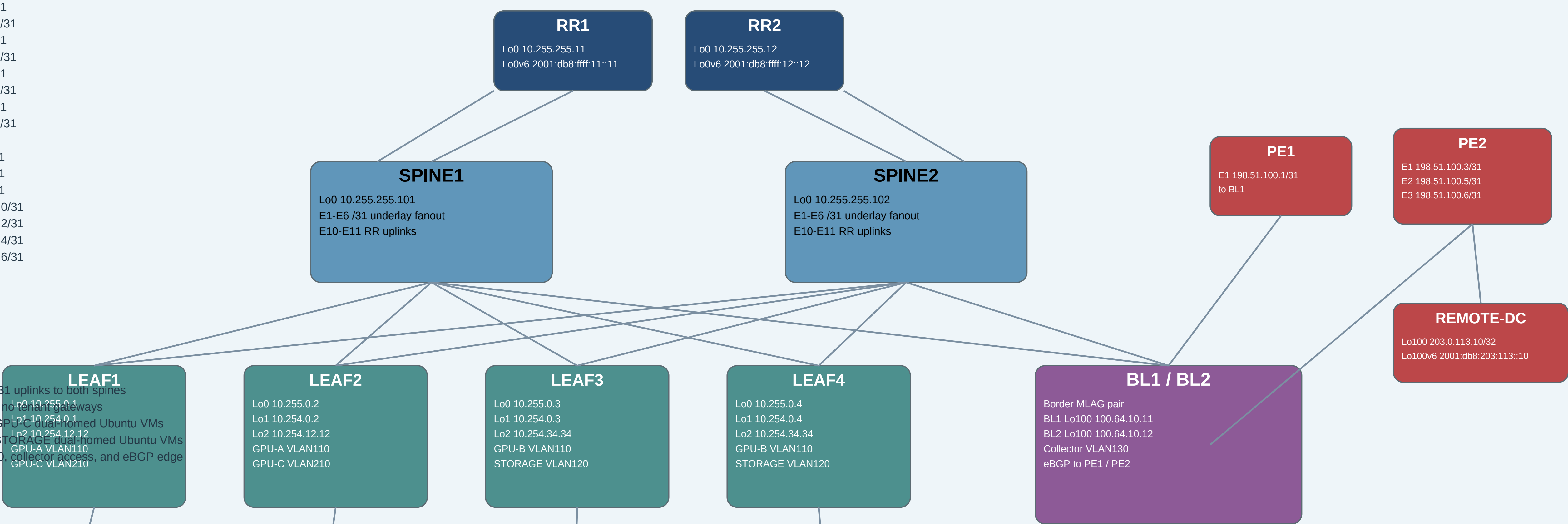
Underlay / 31 Mapping

Core Fabric / Underlay mapping, EVPN/VXLAN roles, and border connectivity

LEAF1 E1 10.0.0.0/31 <=> SPINE1 E1 10.0.0.1/31
LEAF1 E2 10.0.0.12/31 <=> SPINE2 E1 10.0.0.13/31
LEAF2 E1 10.0.0.2/31 <=> SPINE1 E2 10.0.0.3/31
LEAF2 E2 10.0.0.14/31 <=> SPINE2 E2 10.0.0.15/31
LEAF3 E1 10.0.0.4/31 <=> SPINE1 E3 10.0.0.5/31
LEAF3 E2 10.0.0.16/31 <=> SPINE2 E3 10.0.0.17/31
LEAF4 E1 10.0.0.6/31 <=> SPINE1 E4 10.0.0.7/31
LEAF4 E2 10.0.0.18/31 <=> SPINE2 E4 10.0.0.19/31
BL1 E1 10.0.0.8/31 <=> SPINE1 E5 10.0.0.9/31
BL1 E2 10.0.0.20/31 <=> SPINE2 E5 10.0.0.21/31
BL2 E1 10.0.0.10/31 <=> SPINE1 E6 10.0.0.11/31
BL2 E2 10.0.0.22/31 <=> SPINE2 E6 10.0.0.23/31
RR1 E1 10.0.100.1/31 <=> SPINE1 E10 10.0.100.0/31
RR1 E2 10.0.100.3/31 <=> SPINE2 E10 10.0.100.2/31
RR2 E1 10.0.100.5/31 <=> SPINE1 E11 10.0.100.4/31
RR2 E2 10.0.100.7/31 <=> SPINE2 E11 10.0.100.6/31

Fabric Role Notes

RR1/RR2: EVPN route reflectors with dedicated /31 uplinks to both spines
SPINE1/SPINE2: underlay transit and RR fanout; no tenant gateways
LEAF1/2: POD1 MLAG pair serving GPU-A and GPU-C dual-homed Ubuntu VMs
LEAF3/4: POD2 MLAG pair serving GPU-B and STORAGE dual-homed Ubuntu VMs
BL1/BL2: border MLAG pair for services VLAN130, collector access, and eBGP edge

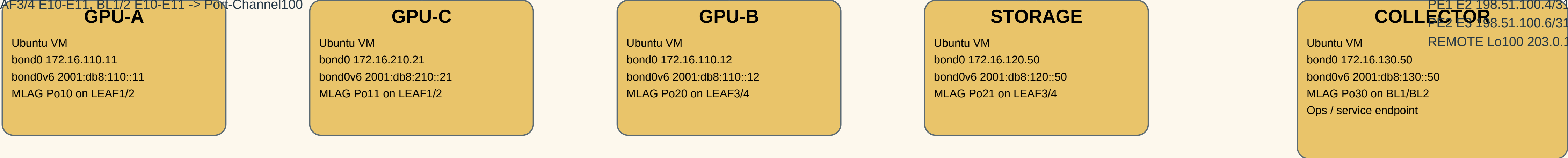


Host / Edge Zone

Host Attachment Mapping

Ubuntu VM endpoints, MLAG host attachment, dual-stack tenant services, and external edge mapping

GPU-A eth0 -> LEAF1 E20 | eth1 -> LEAF2 E20 | Port-Channel10 | VLAN110
GPU-C eth0 -> LEAF1 E21 | eth1 -> LEAF2 E21 | Port-Channel11 | VLAN210
GPU-B eth0 -> LEAF3 E20 | eth1 -> LEAF4 E20 | Port-Channel20 | VLAN110
STORAGE eth0 -> LEAF3 E21 | eth1 -> LEAF4 E21 | Port-Channel21 | VLAN120
COLLECTOR eth0 -> BL1 E20 | eth1 -> BL2 E20 | Port-Channel30 | VLAN130
Peer-links: LEAF1/2 E10-E11, LEAF3/4 E10-E11, BL1/2 E10-E11 -> Port-Channel100



Loopback / VTEP Mapping

POD1 shared VTEP: LEAF1 Lo2 + LEAF2 Lo2 = 10.254.12.12
POD2 shared VTEP: LEAF3 Lo2 + LEAF4 Lo2 = 10.254.34.34
BORDER shared VTEP: BL1 Lo2 + BL2 Lo2 = 10.254.78.78
LEAF1 Lo1 10.254.0.1 | LEAF2 Lo1 10.254.0.2
LEAF3 Lo1 10.254.0.3 | LEAF4 Lo1 10.254.0.4
BL1 Lo1 10.254.0.11 | BL2 Lo1 10.254.0.12

External Edge Mapping

BL1 E30 198.51.100.0/31 -> PE1 E1 198.51.100.1/31
BL2 E30 198.51.100.2/31 -> PE2 E1 198.51.100.3/31
PE1 E2 198.51.100.4/31 -> PE2 E2 198.51.100.5/31
PE2 E3 198.51.100.6/31 -> REMOTE E1 198.51.100.7/31
REMOTE Lo100 203.0.113.10/32 | Lo100v6 2001:db8:203:113::10/128

Dual-Stack VLAN / VNI / Gateway Mapping

VLAN110 -> VNI10110 -> VRF AI_FABRIC
172.16.110.0/24 GW 172.16.110.1 | 2001:db8:110::/64 GW 2001:db8:110::1
VLAN120 -> VNI10120 -> VRF AI_FABRIC
172.16.120.0/24 GW 172.16.120.1 | 2001:db8:120::/64 GW 2001:db8:120::1
VLAN130 -> VNI10130 -> VRF AI_SERVICES
172.16.130.0/24 GW 172.16.130.1 | 2001:db8:130::/64 GW 2001:db8:130::1
VLAN210 -> VNI10210 -> VRF TENANT_B
172.16.210.0/24 GW 172.16.210.1 | 2001:db8:210::/64 GW 2001:db8:210::1